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CDS505: DATA VISUALISATION & VISUAL ANALYTICS

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GROUP PROJECT

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1.0 Introduction

1.1 Background

The Life Expectancy (WHO) dataset serves as a critical resource for understanding the complex web of factors that determine human longevity. In this data-driven era, public health policy relies heavily on data in decision-making. This dataset enables analysts to move beyond simple mortality statistics and explore the causal relationships between a nation's health, economic status, and social infrastructure. By aggregating data from 179 countries, the dataset provides a unique opportunity to identify the key determinants of longevity and formulate actionable strategies for improvement. This dataset is essential for application in multidisciplinary frameworks that integrate epidemiology, development economics, and predictive analytics to drive systemic global health improvements.

Link to website hosted on github pages : <https://taneesheng.github.io/CDS505-Project/>

1.2 Big Picture: The Real-World Problem

The core real-world problem addressed by this analysis is the profound disparity in global health outcomes and the disconnection between health indicators, socioeconomic factors and actual longevity. In a data-driven era, public health policy relies heavily on empirical evidence to make decisions, yet significant gaps remain between intervention planning and health outcomes.

The specific real-world challenges this project addresses include:

Extreme Global Inequality

The analysis highlights a massive demographic crisis, exemplified by a 40-year life expectancy gap between top-performing nations like Japan (approximately 83 years) and struggling nations like Lesotho (approximately 46 years). This disparity represents a documented global health crisis that requires immediate identification of structural causes.

Allocative Efficiency

A critical problem for governments is determining allocation efficiency. GDP serves as a proxy for a nation's capacity to mobilize resources, but it does not guarantee health. Policymakers need to understand why some nations significantly underperform their regional economic peers, isolating inefficiencies such as inequitable access or poor public hygiene rather than attributing deficits solely to insufficient funding.

Strategic Aid Deployment

International bodies often struggle with the precise identification of aid efficacy. The goal of the big picture is to move from general financial aid to targeted "Health Diplomacy," where data suggests whether a country needs direct capital or technical medical expertise based on whether their binding constraints are financial or structural.

1.3 Dataset Description and Preparation

The data spans a critical development period from 2000 to 2015. While the primary health indicators were sourced from the World Health Organization (WHO) and United Nations (UN). The original versions of this dataset faced the issue of missing values and glaring inaccuracies that hindered reliable analysis. To create this robust version, extensive cleaning and validation were performed.

To ensure data integrity and statistical validity, the dataset underwent a rigorous cleaning process where economic and demographic variables, such as GDP and Population, were

performed with cross-referencing and corrected against official World Bank records. Rather than discarding the incomplete entries, missing data points were carefully imputed by applying the nearest 3 year averages or regional estimates, while countries missing more than four distinct indicators were excluded entirely. This comprehensive quality control resulted in a refined and robust dataset of 2,864 rows and 21 columns, providing an accurate socioeconomic context for analysis. The dataset is structured to provide a holistic view of national health, categorised into three primary domains, which include health and mortality, immunization and prevention and socioeconomic factors.

1.4 Target Audience

This analysis is designed to serve stakeholders who require empirical evidence to bridge the gap between global economic and health intervention planning and health outcomes. Key beneficiaries of this study include below.

Public Health Organization

This analysis helps International bodies and non-governmental organizations (NGOs) focus on epidemiological surveillance and resource mobilization. By utilizing the insights of dataset on immunization coverage and disease prevalence in Hepatitis B, Polio, Measle and Diphtheria, these organizations can identify the red-zoned countries as well as identify the mortality rates which are disproportionately high compared to healthcare spending. This allows for the strategic deployment of vaccination campaigns, medical aid, and sanitation projects to the populations with the highest risk profiles.

Government Policy makers

For government policymakers, the analysis of GDP and life expectancy functions as a critical instrument for evaluating national allocative efficiency. GDP serves as a proxy for a nation's capacity to mobilize resources for health, while life expectancy measures the efficacy of converting those inputs into tangible health outcomes. By integrating these metrics and benchmarking against economic peers, policymakers can evaluate the efficiency of resource allocation. For example, for instance, observing that a nation significantly underperforms its regional economic peers allows policymakers to isolate systemic inefficiencies such as inequitable access or poor public hygiene rather than attributing the deficit solely to insufficient funding.

Furthermore, this framework is integral to foreign policy and the strategic management of international development assistance. It enables the precise identification of aid efficacy and informs 'Health Diplomacy' initiatives. For instance, when engaging with nations that demonstrate high economic output but suboptimal health outcomes, the data suggests that structural inefficiencies, rather than capital deficits, are the binding constraints. Consequently, this supports the deployment of technical medical expertise rather than direct financial aid to maximize diplomatic and humanitarian impact.

Academic Researchers and Data Scientists

This dataset serves as a benchmark for the academic community interested in demography, development economics, and predictive modeling. Researchers and data scientists can use the analysis to strengthen their research methodologies and validate new hypotheses with robust empirical evidence. The comprehensive data on disease prevalence and healthcare investment offers a rich training ground for developing predictive models such as forecasting life expectancy trends or evaluating the long-term impact of immunization programs. Furthermore, the visualization facilitates interdisciplinary studies, allowing scholars to explore

complex correlations between economic indicators (like GDP expenditure) and public health outcomes.

1.5 Related Work

Life expectancy determinants have been the subject of extensive econometric and public health research. A foundational concept in this field is the "Preston Curve," first proposed by Samuel Preston (1975), which demonstrated a non-linear relationship between life expectancy and real per capita income. Preston's work established that while economic growth significantly increases life expectancy in poorer nations, this relationship flattens among wealthier countries, suggesting that after a certain economic threshold, longevity is driven more by social factors and technological advances than by wealth accumulation alone (Preston, 1975).

Recent studies using World Bank and WHO data have expanded on this by isolating specific variables found in this dataset. For instance, Miladinov (2020) analyzed the interaction between socioeconomic development and life expectancy in transition economies, finding that while GDP remains a strong predictor, its impact is heavily mediated by the quality of the health system and infant mortality rates. This supports the inclusion of Infant deaths and GDP as critical features for analysis.

Furthermore, the role of education in this dataset as 'Schooling' has been identified as potentially the single strongest predictor of life expectancy. Lutz and Kebede (2018) argued that better education drives health cognition, allowing populations to make better lifestyle choices and utilize healthcare more effectively. Their findings suggest that once education is accounted for, the direct link between income and health weakens, highlighting the importance of the 'Schooling' variable in predictive modeling. Conversely, studies on healthcare expenditure often show mixed results. For example, outcomes can vary significantly based on whether spending is public or private, a nuance partly captured by the dataset's Percentage expenditure and Total expenditure fields (Nazrul et al., 2015).

2. Data Abstraction

2.1 Dataset Type

The dataset is a **flat multivariate table dataset**, which is one of the most common data forms used in information visualisation and visual analytics. The dataset is flat because all observations are stored in a single relational table without nested structures, hierarchies, networks, or links between rows. Each row represents an independent observational unit, specifically a country-year combination, with no parent-child or graph relationships between items. At the same time, the dataset is multivariate because each country-year observation is described by a large number of attributes capturing different dimensions of public health, socio-economic conditions, and demographics, including outcome variables such as life expectancy and mortality rates, explanatory variables such as GDP, schooling, and immunisation coverage, as well as categorical descriptors such as region and development status. This high dimensionality enables complex analytical tasks such as correlation analysis, trend comparison, outlier detection, and multivariate pattern discovery, which are central goals of visual analytics and data storytelling. The reason choosing this dataset is that the combination of a flat structure with rich multivariate attributes should make the dataset particularly suitable for exploratory visual analytics and storytelling.

2.2 Data Types

The attributes in the dataset can be classified into nominal, ordinal, and quantitative data types, following standard data abstraction principles in information visualisation. Categorical (Nominal) attributes describe qualitative groupings or labels and are primarily

used for grouping, filtering, and comparison, for examples in this dataset include Country, Region, and Economy Status (Developed/Developing). Categorical (Ordinal) attributes have a meaningful sequence but not necessarily a fixed numeric distance between values, with Year being the most important ordered attribute in this dataset to enable temporal analysis and trend exploration. Then, quantitative attributes represent numerical measurements with meaningful magnitudes and ratios, such as Life Expectancy, Adult Mortality, GDP per Capita, Schooling, and Immunisation Coverage. These quantitative variables are suitable for precise visual encodings, including position, length, and colour intensity. Together, these attribute types can support the analytical tasks, from temporal trend analysis and cross-country comparison to multivariate correlation exploration. The table below presents the full list of attributes in the dataset and their corresponding data types.

Table 1 : Data attribute of Life Expectancy (WHO) dataset

Attribute	Attribute Type	Description
Country	Categorical (Nominal)	Name of the country
Region	Categorical (Nominal)	Geographical region of the country
Year	Quantitative	The calendar year of the data record
Infant_deaths	Quantitative	Number of infant deaths per 1000 population
Under_five_deaths	Quantitative	Number of deaths of children under five years of age per 1000 population
Adult_mortality	Quantitative	Probability of dying between 15 and 60 years of age per 1000 population
Alcohol_consumption	Quantitative	Recorded alcohol consumption per capita (15+) in litres of pure alcohol
Hepatitis_B	Quantitative	Hepatitis B (HepB3) immunization coverage among 1-year-olds (%)
Measles	Quantitative	Measles (MCV) immunization coverage among 1-year-olds (%)
BMI	Quantitative	Average Body Mass Index of the entire population
Polio	Quantitative	Polio (Pol3) immunization coverage among 1-year-olds (%)
Diphtheria	Quantitative	Diphtheria tetanus toxoid and pertussis (DTP3) immunization coverage among 1-year-olds (%)
Incidents_HIV	Quantitative	Incidents of HIV per 1000 population aged 15-49
GDP_per_capita	Quantitative	Gross Domestic Product per capita (in USD)

Population_mln	Quantitative	Total population in millions
Thinness_ten_nineteen_years	Quantitative	Prevalence of thinness among adolescents aged 10 to 19 years (%)
Thinness_five_nine_years	Quantitative	Prevalence of thinness among children aged 5 to 9 years (%)
Schooling	Quantitative	Average number of years of schooling
Economy_status_Developed	Categorical (Nominal)	Binary indicator if the country status is 'Developed' (1=Yes, 0=No)
Economy_status_Developing	Categorical (Nominal)	Binary indicator if the country status is 'Developing' (1=Yes, 0=No)
Life_expectancy	Quantitative	Life expectancy at birth in years

2.3 Derived Dataset Types

Derived Geometry Dataset: By applying geographic encoding to the Country attribute, the flat multivariate table is expected to be transformed into a derived spatial (geometry) dataset. Although the dataset does not explicitly include latitude and longitude coordinates, country names can be automatically geocoded using built-in geographic reference data available in modern visualisation systems like Tableau. This process converts nominal attributes into spatial geometries to enable countries to be represented as geographic regions on a world map, where spatial position acts as a primary visual encoding. Spatial and map-based visualisations are a fundamental form of information visualization, as they leverage positional encodings to support geographic comparison, pattern recognition, and the identification of spatial disparities, particularly in multivariate socio-economic and public health datasets (Heer et al., 2010). By locating life expectancy values within geographic space, the derived spatial dataset is expected to enhance interpretability and support visual analytics and data storytelling.

Derived Temporal Dataset: Although the raw dataset is structured as a static flat table, the presence of the sequential quantitative attribute, like Year, allows for its transformation into a Derived Temporal Dataset. By indexing the data on the Year column (spanning 2000 - 2015), we can let the dataset move beyond a simple collection of independent records to become a structured time-series. In this derived form, the Country attribute serves as the key entity, while Year provides the temporal axis to enable the analysis of longitudinal trends rather than just cross-sectional snapshots. Then, this derivation is also critical for visual analytics as it supports tasks related to temporal evolution, such as identifying trajectories of health improvement or detecting periods of stagnation. Without explicitly treating Year as a temporal dimension, our analysis would be limited to static comparisons that fail to capture the dynamic nature of global health development (Aigner et al., 2023).

2.4 Derived Data Types

To specifically explore the relationship between changes in life expectancy and each of the underlying drivers, we systematically derived different types of analytical attributes:

Geographic Hierarchy attributes: Although Region and Country exist as separate nominal attributes in the raw dataset, they do not explicitly define a tree structure. To enable multi-scale visual analysis, a Geographic Hierarchy is derived by structurally linking these attributes from

the World to Regions to Countries. This transformation converts the flat tabular data into a nested structure to allow for Hierarchical Aggregation. In this process, quantitative values like Life Expectancy are automatically averaged at the Region level before being displayed at the Country level. This derived structure is a prerequisite for Drill-Down interactions, which allow users to follow Shneiderman's Information Seeking Mantra by starting with a global overview of regional aggregation and zooming in to specific details like country-level data only when requested. (Elmqvist & Fekete, 2010)

Binning attributes: In order to highlight differences within factors and present clear visual comparisons, the continuous variables of 'Thinness_Bin' and 'BMI_Bin' were derived by binning as 'low,' 'medium,' or 'high,' based on key health and demographic thresholds. This classification allows us to transform the data into groups of clear relevance, thus allowing us to visualize and compare differences and patterns in life expectancy between groups of countries with different nutritional status.

Normalized attributes: By comparing and analyzing the visualizations of socioeconomic and health factors, the factors most strongly associated with life expectancy, such as 'Infant Mortality', 'Adult Mortality', and 'Schooling', were selected. To eliminate dimensional differences and achieve cross-factor comparability, as well as to provide a more intuitive comparison of the relative strengths and weaknesses of the indicators across years for the countries with the largest increases in life expectancy, these key factors were normalized using its global minimum and maximum values and the following variables were derived: 'Norm_Infant_Mortality,' 'Norm_Adult_Mortality,' 'Norm_Schooling,' 'Norm_Diphtheria,' 'Norm_Polio,' 'Norm_GDP,' and 'Norm_BMI.'

Change attributes: To gain deeper insight into the drivers behind shifts in life expectancy, we first derived 'Life_Expectancy_Change' as our core dependent variable, calculated as its growth rate between 2000 and 2015. To investigate the relationship between changes in various factors and life expectancy, we then constructed a core calculation field ('Factor_Growth_Rate') that dynamically calculates the compound rate of change over the same period (2000-2015) of seven key factors based on analytical requirements. In the visualization, this dynamic field is used on the horizontal axis to compare the magnitude of change of different factors among the countries with the fastest-growing life expectancy.

3.0 Task Abstraction

Task abstraction transforms domain-specific questions into generalised analytical operations and addresses them with visualizations. Following Munzner's framework, we identify four primary tasks that enable users to understand the life expectancy across regions and countries, and the determinants of life expectancy. Each task is specified into its action and target, which represent what the user does and the data aspects being examined.

3.1 Task 1: Global Health Inequality

Action: Discover, Locate, Compare, Identify

Target: Spatial Data, Distribution, Extremes, Trends

This is to investigate global disparities in health outcomes to understand where inequality is most profound. The analysis begins by using spatial data to locate geographic regions with critical health deficits to enable users to detect contiguous clusters of low life expectancy. This approach leverages the power of maps to highlight spatial autocorrelation to enable users to see how data values are distributed across geographic space. To move beyond simple averages, the task requires comparing regional distributions to assess the variance and inequality within specific continents. The use of box plots is essential as they provide a robust summary of distribution characteristics, such as medians and interquartile ranges, to allow for

efficient comparison between multiple groups while hiding the noise of individual data points (Healy, 2019). Simultaneously, users need to identify and rank the specific nations at the extremes of the spectrum, the top and bottom performers, to quantify the absolute gap in longevity. Finally, to determine if global health initiatives are effective, the task involves analysing temporal trends. Line charts are needed here because they are the standard encoding for time-series data to enable users to instantly perceive the direction and rate of change over the 15 years (Dougherty, 2021).

Analytical Questions

- Which geographic regions are clustered with the lowest life expectancy?
- What is the magnitude of the gap between the top 5 and bottom 5 countries?
- How does the variance (inequality) of life expectancy differ between regions?
- Has the life expectancy gap between regions narrowed or widened from the year 2000 to 2015?

Table 2 : Visualisation Type and Purpose of Task 1

Visualisation Type	Purpose
Choropleth Map	Locate spatial clusters of high and low life expectancy (Global Distribution) to detect geographic patterns.
Bar Chart	Identify the extremes (Top and Bottom 5 Countries) to quantify the maximum disparity.
Box Plot	Compare the spread and median of life expectancy (Region vs. Life Expectancy) to reveal inequality within regions.
Line Chart	Analyze trends over time (Life Expectancy by Year) to track progress and temporal changes.

3.2 Task 2: Socioeconomic Factors (GDP & Schooling)

Action: Discover, Locate, Identify, Compare

Target: Correlation, Outliers, Distribution, Spatial Data

In the socioeconomics task abstraction, the focus is on exploratory analysis to discover patterns between socioeconomic indicators and quality-of-life outcomes, transforming the user from a passive viewer to an active analyst. To examine whether economic prosperity predicts longevity, the task enables users to analyse correlations between quantitative variables while simultaneously querying for outliers, such as wealthy nations with low life expectancy. This dual focus is supported by the classification of characterising trends and identifying outliers as the two fundamental targets for scatter plot idioms (Munzner, 2014). Beyond trends, the analysis requires quantifying the educational gap between developed and developing economies through direct comparison of aggregated attributes, and assessing economic inequality by analysing the full distribution of wealth across regions rather than misleading averages (Wilke, 2019). Finally, recognising the inherent geographic nature of the data, the design mandates a spatial component to search and locate specific clusters of health outcomes, allowing users to verify if neighbouring regions share similar characteristics.

Analytical Questions

- Does higher GDP per capita lead to longer life expectancy?

- What is the education gap between developed and developing nations?
- How does the distribution of wealth vary across different regions?
- Which geographic regions suffer from the lowest life expectancy?

Table 3 : Visualisation Type and Purpose of Task 2

Visualisation Type	Purpose
Scatter Plot	Discover wealth-health trends (GDP vs. Life Expectancy) and identify outliers.
Bar Chart	Compare schooling years (Education Gap) between Developed and Developing countries.
Bar Chart	Compare wealth distributions (GDP per Capita by Region) across regions.
Choropleth Map	Locate geographic clusters and search for spatial health patterns.

3.3 Task 3: Health Indicators (Mortality, Vaccination, BMI & Thinness)

3.3.1 Mortality

Action: Summarise, Identify, Locate, Compare

Target: Correlation, Trends, Spatial Data, Outliers

The mortality indicator focused on mortality risks to understand the relationship between different life-stage mortality indicators and overall longevity. The central abstract task is to correlate Infant Mortality with Adult Mortality to determine if risks at different life stages are linked or independent. This requires a view that plots these two quantitative variables against each other to reveal linear relationships and outliers, as nations where high infant death rates might not necessarily predict high adult death rates. To assess global progress in reducing death rates, users must monitor trends over time to track whether mortality metrics are declining across decades. This temporal analysis is supported by a summary task to provide users with immediate access to high-level aggregates like KPIs for Life Expectancy, Infant Mortality, and Adult Mortality to establish a global baseline. Then, the analysis needs a spatial component to locate geographic regions with critically high mortality rates to enable users to verify if health crises are regionally clustered or randomly distributed (Wilke, 2019).

Analytical Questions

- Is there a strong correlation between high infant mortality and high adult mortality?
- How have global adult and infant mortality rates changed over the last 15 years?
- Which geographic regions suffer from the highest concentration of adult mortality?
- Are there countries with low infant mortality but unexpectedly high adult mortality (outliers)?

Table 4 : Visualisation Type and Purpose of Task 3(Mortality Dashboard)

Visualisation Type	Purpose
KPI Cards	Summarize single values to provide an immediate high-level overview of global health performance.

Bubble Chart	Correlate vaccination rates and Identify outliers to assess the relationship between immunization and health outcomes.
Multi-Series Line Chart	Monitor temporal trends to compare the progress of multiple immunization programs simultaneously over time.
Choropleth Map	Locate geographic gaps in healthcare to detect spatial patterns of vulnerability.

3.3.2 Vaccination

Action: Summarize, Locate, Compare, Identify

Target: Correlation, Trends, Spatial Data, Outliers

Rather than just observing demographic data, users should be able to evaluate the causal link between specific preventative measures, like Polio vaccination and broader health outcomes. The core analytical engine here is a correlation task that users map the density of immunisation coverage against Life Expectancy to verify if preventative medicine truly acts as a predictor for longer lives. This naturally leads to outlier identification, where nations with high coverage but poor longevity signal that other factors are at play. Then, the sustainability of these programs is assessed by monitoring temporal trajectories. By plotting multiple vaccination types, such as Hepatitis B, Measles, and Diphtheria, on a multi-series timeline, users can compare the maturation of newer health programs against established ones over 15 years (Dougherty, 2021). The analysis grounds these statistics in reality as users locate specific geographic zones where coverage is failing to enable a targeted assessment of where aid is most critical.

Analytical Questions

- Is there a measurable link between higher Polio vaccination rates and increased longevity?
- How does the adoption rate of Hepatitis B vaccines compare to long-standing programs like Diphtheria?
- Which specific regions have failed to achieve the target threshold for Polio immunization?
- Are there nations with high vaccination coverage that still suffer from low life expectancy?

Table 5 : Visualisation Type and Purpose of Task 3(Vaccination Dashboard)

Visualisation Type	Purpose
Bubble Chart	Correlate preventative care and identify efficacy gaps where high coverage does not yield high longevity.
KPI Cards	Summarize key performance indicators to establish a baseline for healthcare success.
Multi-Series Line Chart	Monitor and compare the temporal progress of different immunisation programs over decades.
Choropleth Map	Locate spatial loss in healthcare access to pinpoint vulnerable populations.

3.3.3 BMI and Thinness

Action: Summarise, Identify, Compare, Locate

Target: Correlation, Distribution, Trends, Spatial Data

It is important to tackle the nutritional determinants of health, like investigating how Body Mass Index (BMI) and thinness rates influence global longevity. This investigation is the need to correlate average national BMI against Life Expectancy to allow users to assess whether higher nutritional markers serve as a proxy for economic development and longer life, or if they introduce new health risks. This analysis goes beyond simple averages by requiring users to compare vulnerability across different life stages, specifically showing thinness rates in children (5 - 9 years) versus adolescents (10 - 19 years) to pinpoint critical intervention windows. To contextualise these findings, the dashboard enables users to monitor the gradual shift in global body composition over time, tracking whether the world is collectively trending towards obesity or if malnutrition persists. With these metrics in reality, the spatial component can allow users to locate regional differences and then distinguish between nations facing crises of undernutrition versus those managing lifestyle-related health burdens.

Analytical Questions

- How does the prevalence of childhood thinness (ages 5 - 9) compare to adolescent thinness (ages 10 - 19)?
- Does a higher average BMI correlate with higher or lower life expectancy globally?
- Which world regions exhibit the lowest average BMI, indicating potential undernutrition?
- Have global thinness rates declined over the past 15 years as BMI averages have risen?

Table 6 : Visualisation Type and Purpose of Task 3(BMI Dashboard)

Visualisation Type	Purpose
Bubble Chart	Correlate nutritional status like BMI vs. Life Expectancy, and assess if body mass serves as a valid predictor of health outcomes.
KPI Cards	Summarise nutritional benchmarks to establish a global baseline for physical health.
Multi-Series Line Chart	Monitor temporal shifts such as BMI and Thinness Trends to track the changing landscape of global nutrition over decades.
Choropleth Map	Locate spatial clusters of high or low body mass to distinguish regions of scarcity from regions of excess.

3.4 Task 4: Life Expectancy Improvement

Action: Identify, Compare, Locate, Discover

Target: Trends, Correlation, Extremes, Spatial Data

This life expectancy improvement is to identify which nations have achieved the most significant improvements in life expectancy and to determine the underlying factors fueling this growth. Users must rank top-performing countries to isolate success stories from the global average. To understand why these improvements occurred, the dashboard facilitates a complex correlation task by plotting the change in GDP per capita against the change in Life

Expectancy to see if economic acceleration is the primary catalyst for health recovery. Furthermore, the analysis needs a comparative assessment of influence, by normalising disparate variables such as mortality, immunisation, schooling, and GDP onto a single scale, users can visually weigh which socioeconomic factor has historically exerted the strongest pressure on longevity. The task also involves locating clusters of rapid improvement, then highlighting regions that may be recovering from historical lows.

Analytical Questions

- Which specific countries have seen the largest percentage increase in life expectancy over the last 15 years?
- Is there a direct correlation between rapid economic growth (GDP change) and rapid health improvement?
- Among the "Top 10" improving nations, which factor (Schooling vs. GDP) shows the strongest normalised influence?
- Are the regions with the highest improvement clustered geographically, or are they isolated success stories?

Table 7 : Visualisation Type and Purpose of Task 4

Visualisation Type	Purpose
Choropleth Map	Locate regions of rapid change to identify geographic zone of recovery.
Bar Chart	Rank top performers to isolate specific countries that outperformed global averages.
Strip Plot	Compare normalised distributions to assess which variable (Mortality, GDP, BMI) has the widest variance and impact.
Bubble Chart	Correlate rates of change to determine if economic booms drive health booms.

4.0 Design Justification

This section justifies the visual design choice for each task and explains the selection of marks, channels and interaction techniques based on visualization principles. The design principles are grounded in Munzner's (2014) effectiveness rankings for visual channels and the expressiveness principle. It states that visual encodings should match the characteristics of the data being represented (Munzner, 2014).

4.1 Marks and Channels Overview

Visual encoding involves mapping data attributes to visual channels through marks. The effectiveness of different channels varies based on the data type being encoded. For example, position is the most effective channel in quantitative data, followed by length, angle and area. For categorical data, spatial region and color hue are most effective (Munzner, 2014).

Table 8 : Marks and Channels Used

Mark Type	Best Use Case	Channel Used	Our Application
Point	Individual Items	Position, Color, Size	Scatter plots, Bubble charts
Line	Connections, Trends	Position, Slope	Time-series line charts
Area	Regions, Quantities	Size, Color saturation	Choropleth maps, KPI cards
Bar (Line)	Comparisons	Length, Position	Bar charts, Rankings

4.2 Task 1: Global Health Inequality

Figure 1 : Dashboard for Task 1- Global Health Inequality

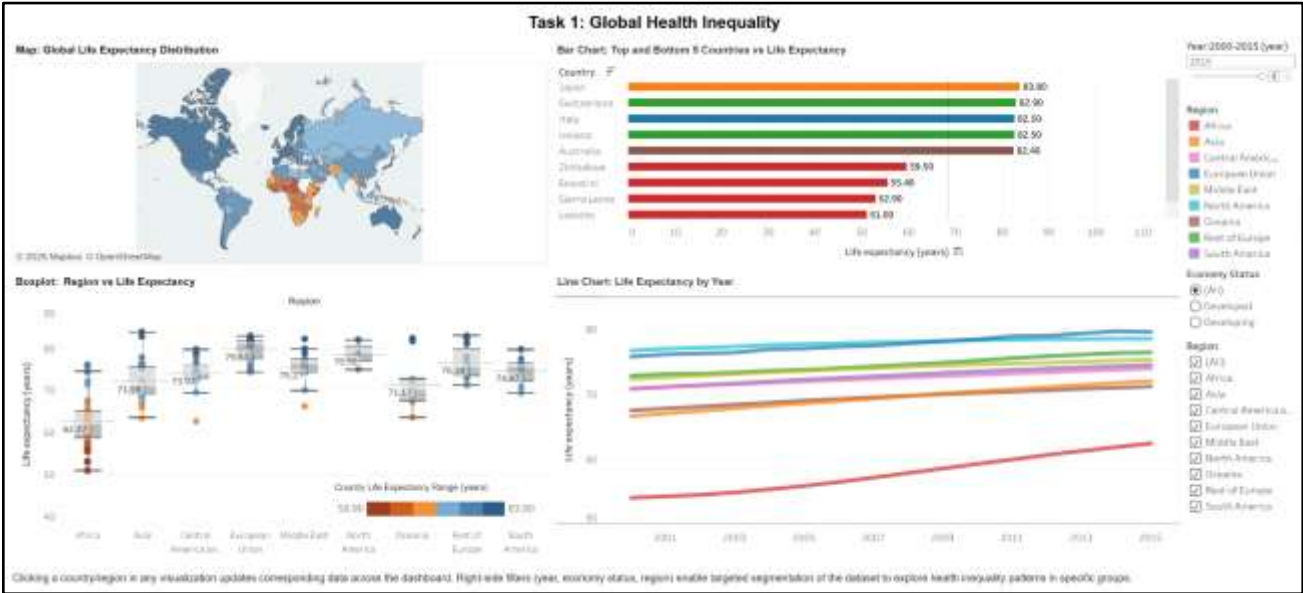


Table 9 : Design Justification for Visualization 1.1 - Choropleth Map (Global Life Expectancy Distribution)

Design Element	Justification
Idiom	Choropleth Map, multiple geographic regions filled with color based on data values
Mark	Area or country polygons to represent geographic regions as continuous spatial units
Channel: Position	Geographic coordinates encode spatial location, leveraging innate human ability to recognise geographic patterns
Channel: Color Saturation	Sequential color scale (light orange to dark red) encodes life

	expectancy magnitude. Darker colors means lower life expectancy, creating a 'danger' association
Interaction	Hover tooltips reveal exact values, Region filter enables focus on specific continents
Design Rationale	Choropleth maps are optimal for showing geographic distributions of a single quantitative variable. Users can instantly identify spatial clusters, such as the African region's low life expectancy, without reading individual values.

Table 10 : Design Justification for Visualization 1.2 - Bar Chart (Top 5 and Bottom 5 Countries vs. Life Expectancy)

Design Element	Justification
Idiom	Horizontal Bar Chart with diverging layout where top performers at above, bottom performers at below
Mark	Line (bar) , length is the most accurate channel for comparing quantitative values after position
Channel: Length	Bar length encodes life expectancy value, enabling precise comparison between countries
Channel: Position	Vertical position encodes rank where sorted order reveals hierarchy without additional encoding
Channel: Color Hue	Categorical color distinguishes top performers (green/blue) from bottom performers (orange/red)
Design Rationale	Bar charts with a common baseline enable accurate length comparison. Showing extremes quantifies the 40-year gap (Japan 83 vs Lesotho 46) more precisely than the map alone.

Table 11 : Design Justification for Visualization 1.3 - Box Plot (Region vs Life Expectancy)

Design Element	Justification
Idiom	A box plot (Box-and-Whisker) shows statistical summary visualization and distribution characteristics
Mark	Composite of Box (area) shows IQR, Line (whiskers) shows range and Points show outliers
Channel: Position	Vertical position encodes life expectancy values; horizontal position separates regions
Channel: Length	Box height encodes interquartile range (IQR), revealing variance within each region
Channel: Color	Color hue distinguishes regions and saturation encodes average life expectancy
Design Rationale	Box plots efficiently summarize distributions using five statistics (min, Q1, median, Q3, max). This reveals that Africa has both the lowest median and highest variance, while Europe has high median with low variance.

Table 12 : Design Justification for Visualization 1.4 - Line Chart (Life Expectancy by Year)

Design Element	Justification
Idiom	Multi-series Line Chart , multiple lines on shared axes for trend comparison
Mark	Line (connection mark) shows continuity and trends over sequential data
Channel: Position X	Horizontal position encodes Year (ordinal), progressing left-to-right following reading convention
Channel: Position Y	Vertical position encodes life expectancy (quantitative), with higher = better
Channel: Color Hue	Distinct hues differentiate 9 regions; colorblind-safe palette used
Interaction	Region filter checkbox allows focus on specific regions where hover shows exact values
Design Rationale	Line charts are optimal for time-series data as connection marks emphasize trends and slopes. Users can see that all regions improved, but Africa's steeper slope indicates faster catch-up progress.

4.3 Task 2: Socioeconomic Factors

Figure 2 : Dashboard for Task 2 - Socioeconomic Factors

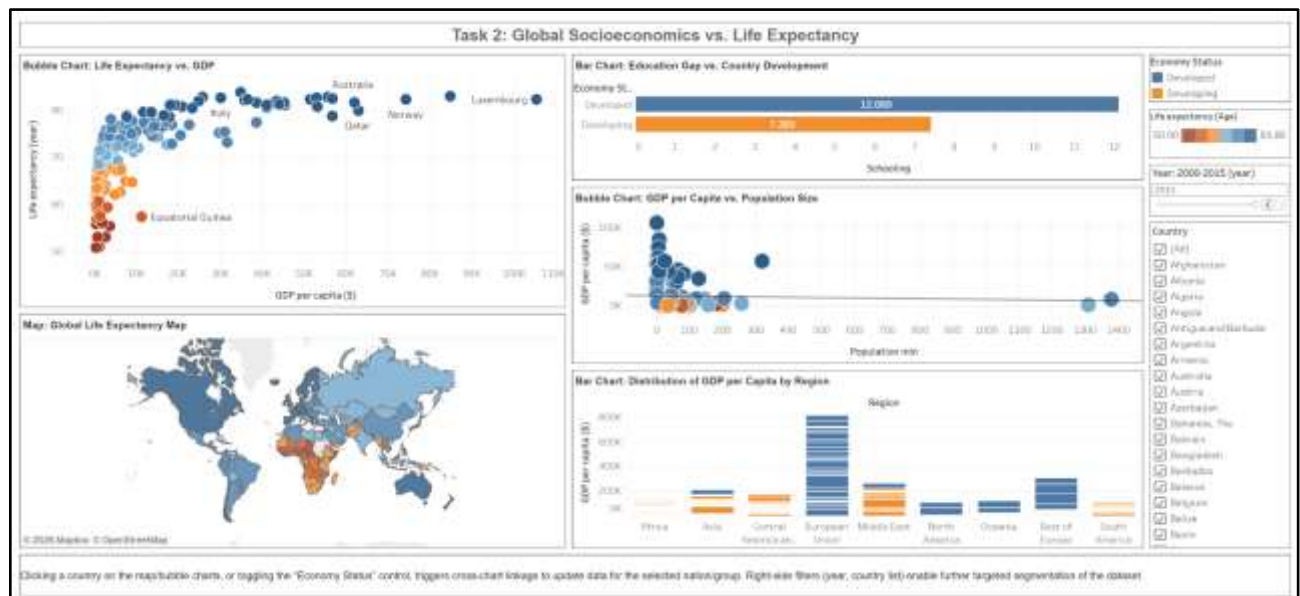


Table 13 : Design Justification for Visualization 2.1 - Bubble Chart (Life Expectancy vs. GDP)

Design Element	Justification
Idiom	A scatter plot is chosen to show the correlation of two quantitative variables and to help identify outliers.
Mark	Points are used as the primary mark to represent individual country data entries in a 2D space.
Channel: Position X	Position X maps to Avg. GDP per capita to represent the economic independent variable.
Channel: Position Y	Position Y maps to Avg. Life expectancy to show the dependent health outcome variable.
Channel: Color Hue	Color Hue distinguishes different geographic regions to allow for group-based identification within the cluster.
Interaction	Selecting a country point triggers global linked highlighting to track that specific nation across the map and other charts, while hovering displays a tooltip with exact data values.
Design Rationale	This idiom is the most effective for correlating wealth and health, as it reveals the non-linear "Preston Curve" and makes it easy to identify outliers like Equatorial Guinea that deviate from the global trend.

Table 14 : Design Justification for Visualization 2.2 - Bar Chart (Education Gap vs. Country Development)

Design Element	Justification
Idiom	A bar chart is used to support the comparison of discrete categorical groups on a common scale.
Mark	Lines (Bars) act as marks where the geometric length represents the value for each category.
Channel: Length	Length is utilised as it is the most accurate channel for humans to judge quantitative differences.
Channel: Color Hue	Color Hue is applied to emphasise the binary distinction between Developed and Developing economy statuses.
Interaction	Selecting a bar highlights all countries in that economic category across the entire dashboard, facilitating instant group comparisons through linked highlighting.
Design Rationale	A bar chart is used because length on an aligned scale is the most accurate channel for comparing the specific magnitude of the education gap between Developed and Developing nations.

Table 15 : Design Justification for Visualization 2.3 - Bubble Chart (GDP per Capita vs. Population Size)

Design Element	Justification
Idiom	A bubble chart expands on a scatter plot by adding a third dimension of data via mark size.
Mark	Points (Circles) are used as marks where size and position encode different attributes.
Channel: Area	Area encodes population mln to show the relative weight of each country globally.
Channel: Position Y	Position Y represents Avg. GDP per capita to maintain consistency with other economic views.
Channel: Position X	Position X maps to Avg. Population mln to investigate potential dependencies on country size.
Interaction	Users can click individual bubbles to highlight that country's spatial location on the map and its performance in other metrics through the dashboard's global interaction system.
Design Rationale	Using area to encode population adds a third dimension to provide multivariate analysis that prevents small countries from skewing the user's perception of global economic trends.

Table 16 : Design Justification for Visualization 2.4 - Map Chart (Global Life Expectancy Map)

Design Element	Justification
Idiom	A choropleth map is used to locate and discover spatial patterns and regional health disparities.
Mark	Areas (Geographic Shapes) represent nations as marks defined by their spatial boundaries.
Channel: Color Saturation	Color Saturation/Luminance provides an intuitive gradient for quantitative health magnitudes.
Channel: Spatial Region	Spatial Position is determined by real-world coordinates to reveal geographic clustering.
Interaction	The map allows users to click any geographic shape to highlight that country's specific data points in the abstract scatter and bar charts for deeper context.
Design Rationale	The map is essential for Locate tasks to allow users to discover spatial autocorrelation, like clusters of neighbouring countries that share similar health or wealth outcomes.

Table 17 : Design Justification for Visualization 2.5 - Bar Chart (Distribution of GDP per Capita by Region)

Design Element	Justification
Idiom	An aggregate bar chart is used to summarize distributions across broad regional categories.
Mark	Lines (Bars) represent the aggregated average economic value for each region.
Channel: Length	Length effectively shows the massive variance in regional wealth averages.
Channel: Position	Position on a categorical axis allows for easy ranking and identification of regional performers.
Interaction	Clicking a regional bar filters the entire dashboard to show only countries within that region, while the visualization supports ranking regions by wealth.
Design Rationale	This chart summarises and ranks the economic standing of regions to provide a high-level context that explains the health disparities seen in the other visualizations.

4.4 Task 3: Health Indicators

4.4.1 Mortality Factors

Figure 3 : Dashboard for Task 3-1 - Mortality Factors

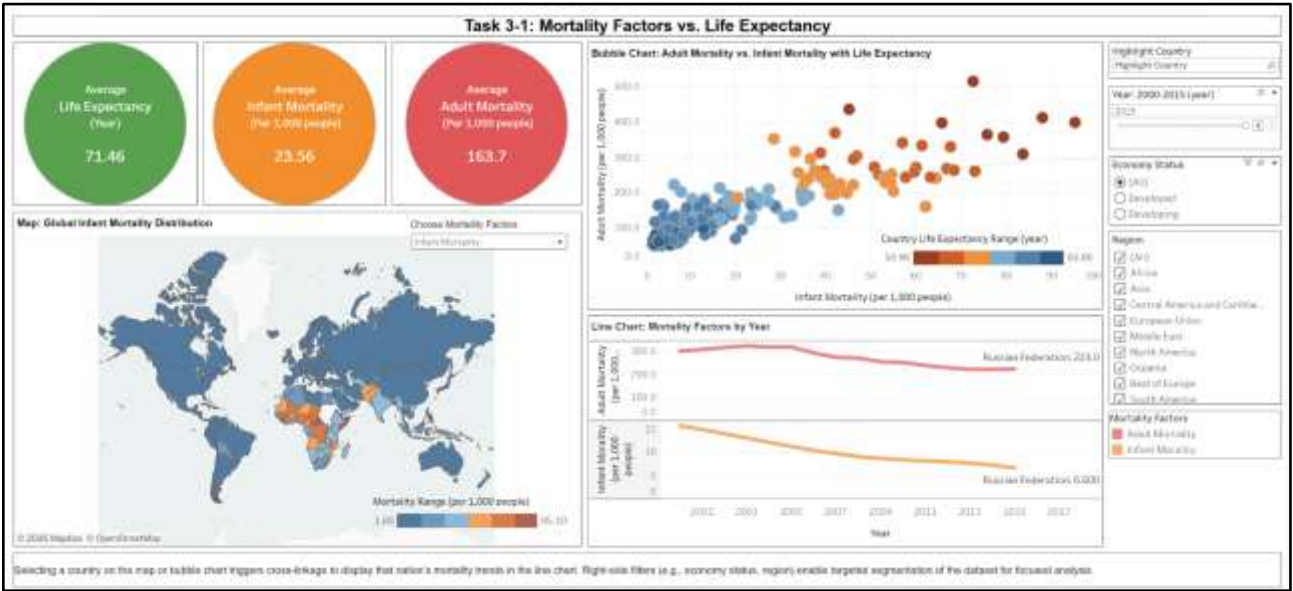


Table 18 : Design Justification for Visualization 3-1.1 - KPI Cards (Global Health Averages)

Design Element	Justification
Idiom	Text-based KPI cards are used to summarize high-level aggregate attributes for immediate reference.
Mark	Text/Enclosing Circles are used as marks to highlight the single most important quantitative values.
Channel: Color Hue	Distinct background colors are applied to each card to help users quickly differentiate between Life Expectancy, Infant, and Adult Mortality.
Interaction	These cards act as dynamic summaries that update automatically whenever a user applies a global filter for Region or Economy Status.
Design Rationale	KPI cards provide the necessary context to summarize the global state of health to give the user a benchmark before they dive into granular spatial or correlation data.

Table 19 : Design Justification for Visualization 3-1.2 - Bubble Chart (Adult Mortality vs. Infant Mortality with Life Expectancy)

Design Element	Justification
Idiom	A bubble chart is used to reveal correlations between two primary quantitative variables while encoding a third via color.
Mark	Points (Circles) are used as marks to represent individual country data in a multidimensional space.
Channel: Position X	Position X maps to Infant Mortality (per 1,000 people) to show the independent variable of early-life survival.
Channel: Position Y	Position Y maps to Adult Mortality (per 1,000 people) to observe how it scales with infant survival rates.
Channel: Color Hue/Saturation	Color encodes the quantitative value of Life Expectancy, transitioning from orange (low) to blue (high) to show dependencies.
Interaction	Clicking any bubble triggers global linked highlighting to isolate that country's mortality history and spatial location across all other dashboard views.
Design Rationale	This idiom effectively supports identifying outliers and assessing the linear dependency between infant and adult health risks to prove how critical early survival is to overall longevity.

Table 20 : Design Justification for Visualization 3-1.3 - Map (Global Adult Mortality Distribution)

Design Element	Justification
Idiom	A choropleth map is used to locate geographic mortality hotspots and discover spatial patterns.
Mark	Areas (Geographic Shapes) represent nations defined by their real-world spatial boundaries.
Channel: Color Saturation	A quantitative color scale represents the magnitude of Adult Mortality, making high-risk regions immediately visible.
Channel: Spatial Region	Spatial position is used to reveal geographic clustering and regional health disparities across continents.
Interaction	Users can click a country on the map to filter and highlight its specific position on the correlation chart and temporal trend.
Design Rationale	The map is essential for Locate tasks to allow users to see spatial autocorrelation where neighboring countries share similar high mortality burdens.

Table 21 : Design Justification for Visualization 3-1.4 - Line Chart (Mortality Factors by Year)

Design Element	Justification
Idiom	A line chart is the standard idiom used to analyze trends and temporal changes in mortality rates over time.
Mark	Lines act as marks that connect individual year data points to emphasize the rate and direction of change.
Channel: Position Y	Position Y maps to the rate of mortality, allowing for a vertical comparison of magnitude over a 15-year period.
Channel: Position X	Position X maps to the temporal attribute (Year), showing progress from 2000 to 2015.
Channel: Color Hue	Color Hue distinguishes between Adult vs. Infant mortality, to prevent visual confusion during comparison.
Interaction	Selecting a country through the global filter or other visualizations updates these lines to show that specific nation's historical health trajectory.
Design Rationale	This idiom supports monitoring global health progress to make it easy to identify whether mortality rates are successfully declining or stagnating over time.

4.4.2 Vaccinations Factors

Figure 4 : Dashboard for Task 3-2 - Vaccination Factors

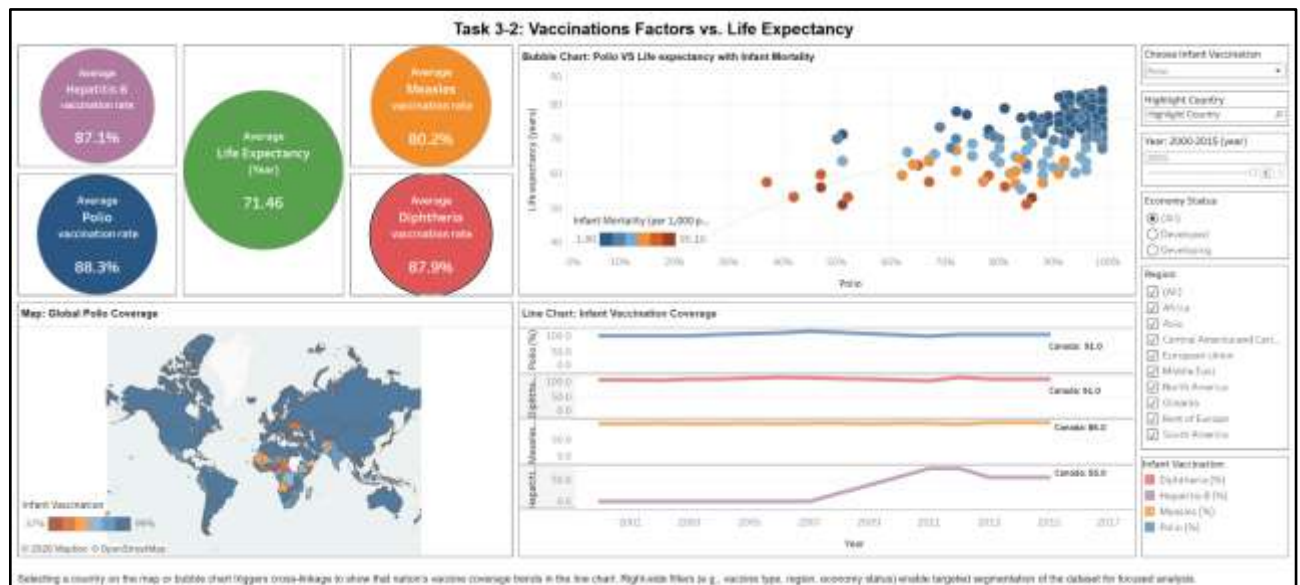


Table 22 : Design Justification for Visualization 3-2.1 - KPI Cards (Average Vaccination Rates)

Design Element	Justification
Idiom	Text-based KPI cards within circles are used to summarize the global state of healthcare interventions.
Mark	Text and enclosing circular areas serve as marks to highlight individual, high-level aggregate values.
Channel: Color Hue	Each circle uses a different hue to associate the aggregate value with its respective detailed trend line in the line chart.
Interaction	These cards act as reactive summaries that update automatically based on any user interaction or global filter applied to the dashboard.
Design Rationale	KPI cards provide the necessary context to summarize global baseline performance, such as the 88.3% Polio rate, before users explore more granular data.

Table 23 : Design Justification for Visualization 3-2.2 - Bubble Chart (Polio VS Life expectancy with Infant Mortality)

Design Element	Justification
Idiom	A bubble chart is chosen to correlate vaccination rates with health outcomes while simultaneously encoding mortality as a visual weight.
Mark	Points (Circles) serve as the primary marks, with each representing a specific country's healthcare profile.
Channel: Position Y	Position X maps to Polio vaccination coverage (Quantitative) to identify it as a potential driver for longevity.
Channel: Position X	Position Y represents Life Expectancy (Quantitative) to measure the dependent outcome of healthcare interventions.
Channel: Color Saturation	Color saturation on a sequential scale represents Infant Mortality rates to highlight multi-dimensional risk.
Interaction	Clicking any point triggers global linked highlighting to reveal that country's spatial coverage on the map and its 15-year vaccination history.
Design Rationale	This idiom is effective for identifying dependencies between preventative care and survival to show that high immunization often correlates with lower infant death and higher life expectancy.

Table 24 : Design Justification for Visualization 3-2.3 - Map (Global Polio Coverage)

Design Element	Justification
Idiom	A choropleth map is used to locate regional immunization gaps and discover spatial patterns in healthcare delivery.
Mark	Areas (Geographic Shapes) represent nations as marks defined by their real-world spatial boundaries.
Channel: Color Saturation	A quantitative gradient scale is to show the density of vaccination coverage across the globe.
Channel: Spatial Region	Spatial Position is determined by geographic coordinates to reveal clusters of vulnerability, particularly in developing regions.
Interaction	Selecting a country on the map updates all other charts in the dashboard to show that specific nation's detailed vaccination stats and health trends.
Design Rationale	The map is essential for Locate tasks to allow users to detect

	spatial autocorrelation where neighboring countries may share similar healthcare infrastructure challenges.
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Table 25 : Design Justification for Visualization 3-2.4 - Line Chart (Infant Vaccination Coverage)

Design Element	Justification
Idiom	A multi-series line chart is the standard idiom used to analyze trends and monitor global health progress over time.
Mark	Lines act as marks connecting year-over-year data points to emphasize the rate and direction of coverage changes.
Channel: Position Y	Position Y represents the percentage of vaccination coverage, allowing for an accurate comparison of magnitude between different programs.
Channel: Position X	Position X represents the temporal dimension, tracking progress from the years 2000 to 2015.
Channel: Color Hue	Distinct color hues are used to separate different vaccination types like Polio, Measles, and Hepatitis B.
Interaction	The line chart dynamically updates through global filters for Region and Economy Status to allow users to see historical trajectories for specific continents.
Design Rationale	This idiom supports monitoring tasks to ensure it easy to identify if global vaccination efforts are successfully scaling up or stagnating.

4.4.3 BMI and Nutrition

Figure 5 : Dashboard for Task 3-3 - Nutrient Factors

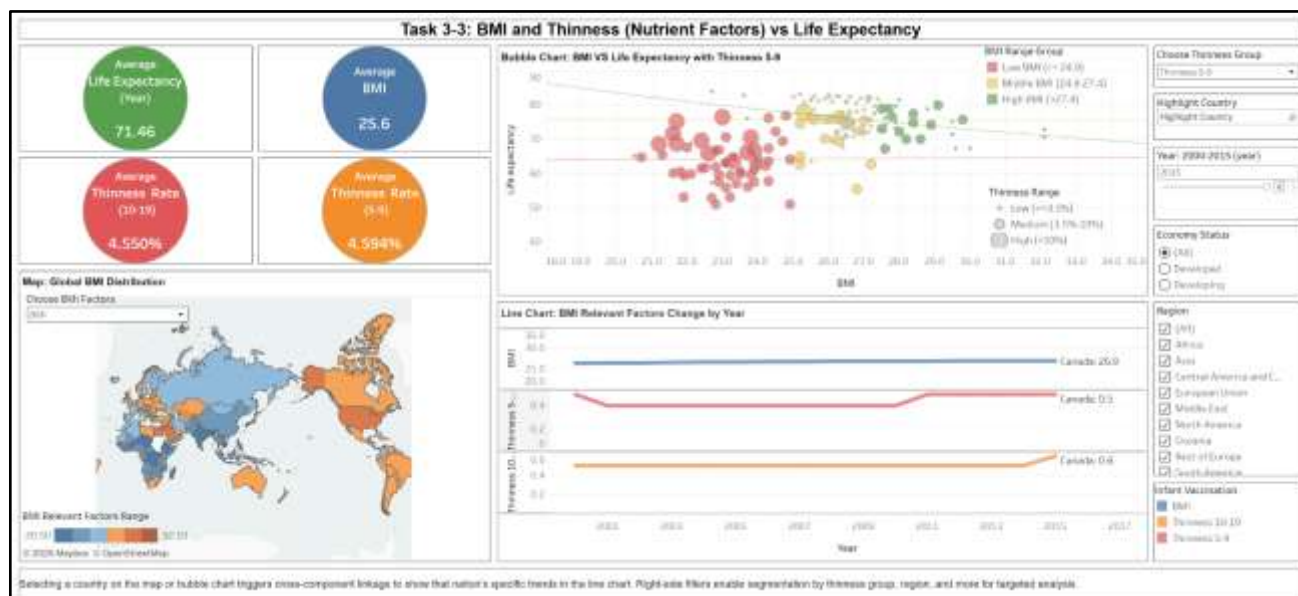


Table 26 : Design Justification for Visualization 3-3.1 - KPI Cards (BMI and Thinness Rate for Different Age Groups)

Design Element	Justification
Idiom	Text-based KPI Cards. Display single and high-level aggregate values.
Mark	Text/Enclosing Circles are used as marks to highlight the single most important quantitative values.
Channel: Color Hue	Distinct background colors are applied to each card to help users quickly differentiate between Life Expectancy, BMI, Thinness Rate (10 - 19) and Thinness Rate (5 - 9).
Interaction	KPI cards with average life expectancy (year), average BMI, average thinness rate (10 - 19) and average thinness rate (5 - 9) will be revealed based on the filtering with economy status, years and regions.
Design Rationale	Immediate Readability with the reduction applied which forces focus on the current state.

Table 27 : Design Justification for Visualization 3-3.2 - BMI VS Life Expectancy with Thinness Rate (Age Group)

Design Element	Justification
Idiom	Bubble Chart, scatter plot where the marks are sized according to a third quantitative variable.
Mark	Point (Scaled Circle). A point (0D) while its Area is scaled to represent data.
Channel: Position X	BMI
Channel: Position Y	Life expectancy
Channel: Area	The size of the bubbles encodes the Thinness Range. Larger bubbles represent a higher thinness rate (more risk). Smaller bubbles represent a lower thinness rate.
Channel: . Color Hue	Green points represent high BMI. Orange points represent middle BMI. Red points represent low BMI.
Interaction	The dotted lines that are colored by group allow users to see the specific correlation within each BMI group, rather than just a global average.
Design Rationale	By splitting the trend lines by color (BMI), allows the viewer to see that the relationship between BMI and Life Expectancy might behave differently for "Low BMI" countries compared to "High BMI" countries.

Table 28 : Design Justification for Visualization 3-3.3 - Map (Global Thinness [Age Group] Distribution)

Design Element	Justification
Idiom	Choropleth Map, filled with color to represent data values. It is the standard approach for visualizing quantitative data across defined geographic boundaries.
Mark	Area (Country polygon). The geometric boundaries are defined by the Latitude and Longitude generated fields.
Channel: Position (Spatial)	Encodes the geographic location of each country.
Channel: Color Hue	Encodes the quantitative metric with BMI Relevant Factors Range (%). Blue hue represents low values (0.10). Orange hue represents high values (27.7). This diverging (or multi-hue) scale is used to highlight severity.

Interaction	Filters for Year, Economy Status, and Region allow the user to isolate specific conditions (e.g., viewing data only for 2011).
Design Rationale	Choropleth Map preserves the neighbor relationships, allowing the viewer to quickly see that this issue is specifically concentrated in South Asia relative to the rest of the hemisphere.

Table 29 : Design Justification for Visualization 3-3.4 - Line Chart (BMI Relevant Factors Change by Year)

Design Element	Justification
Idiom	A multi-series line chart is the standard idiom used to analyze trends and monitor temporal shifts in nutritional status over time.
Mark	Lines act as marks connecting annual data points to emphasize the rate and direction of change in body composition.
Channel: Position Y	Position Y maps to the quantitative values of BMI and thinness rates to allow for a vertical comparison of magnitude across different years.
Channel: Position X	Position X maps to the temporal attribute (Year) to show progress from 2000 to 2015.
Channel: Color Hue	Distinct color hues separate the three different attributes (BMI, Thinness 5-9, and Thinness 10-19) to prevent visual confusion during comparison.
Interaction	The line chart dynamically updates through global filters or interactions with the map and bubble chart to show the historical trajectory of specific countries like Sri Lanka.
Design Rationale	This idiom supports monitoring global nutritional shifts to make it easy to identify if thinness is declining as average BMI rises over time.

4.5 Task 4: Life Expectancy Improvement Factors

Figure 6 : Dashboard for Task 4 - Life Expectancy Improvement Factors

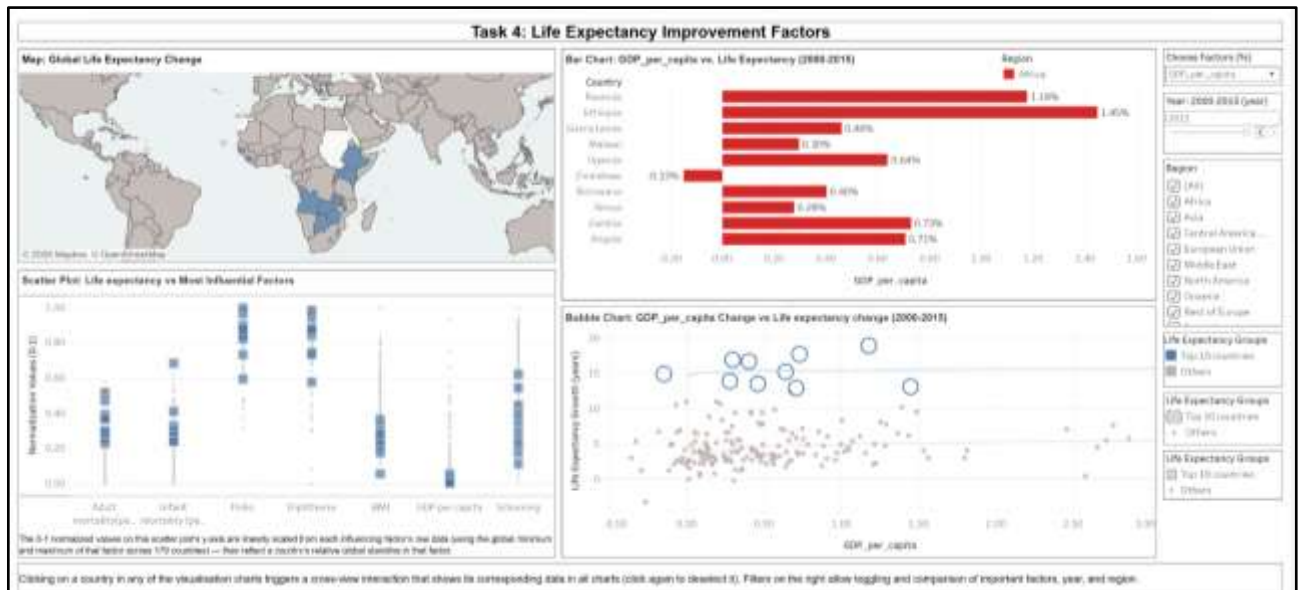


Table 30 : Design Justification for Visualization 4.1 - Choropleth Map (Global Life Expectancy Change)

Design Element	Justification
Idiom	Choropleth Map, the top 10 countries which have large changes are filled with blue color.
Mark	Area (country polygons) represents geographic regions as continuous spatial units
Channel: Position	Determine the horizontal and vertical placement of the country marks on a 2D spatial plane, mirroring the actual physical locations on Earth.
Channel: Color Hue	Countries filled with blue color are used to signify "Top 10 changes," while grey represents others. This is an identity channel used to distinguish between two distinct life expectancy groups.
Interaction	Hover tooltips reveal exact values, Region filter enables focus on specific continents.
Design Rationale	Choropleth Map allows a viewer to recognize patterns of geographic clustering. By highlighting the "Top 10" changes countries, it enables the user to quickly see that the most significant changes are concentrated in Sub-Saharan Africa.

Table 31 : Design Justification for Visualization 4.2 - Bar Chart (Determinant Factors vs Life Expectancy)

Design Element	Justification
Idiom	Horizontal Bar Chart with displaying the top 10 countries which
Mark	Line (bar). Rectangular bars are used as marks to represent the magnitude of the data for each specific country.
Channel: Length	Bar length represents the quantitative value, making it easy to compare magnitudes across countries.
Channel: Position	Vertical position is used for the categorical labels (Country), separating the data into distinct rows.
Channel: Color	Red hue is applied to all marks in this specific view. It likely serves as a consistent visual theme or highlights a specific category like "Region".
Interaction	The dashboard includes a "Choose Factors" parameter dropdown, which are determinants factors (GDP_per_capita, Schooling, BMI, Polio, Diphtheria, Infant Mortality and Adult Mortality) that will reveal the top 10 countries in a bar chart based on the factor selected. Several filters (Year, Economy Status, Region, Country) to refine the data shown.
Design Rationale	The countries are sorted in descending order of the quantitative value. This reduces the cognitive load on the user, as the "top" performers are immediately visible at the top of the list. Including text labels for the values at the end of each bar provides "details-on-demand" without requiring the user to trace the bar back to the x-axis for every single data point.

Table 32 : Design Justification for Visualization 4.3 - Scatter Plot (Life expectancy vs Most Influential Factors)

Design Element	Justification
Idiom	Multivariable Scatter Plot. Displays the distribution of normalised values across several distinct factors (GDP_per_capita, Schooling, BMI, Polio, Diphtheria, Infant Mortality and Adult Mortality).
Mark	Point (Square). Individual data points with squares represent specific country observations within each factor category.
Channel: Position X	Horizontal position is the categorical alignment based on the specific "Measure Name" with factors (GDP_per_capita, Schooling, BMI, Polio, Diphtheria, Infant Mortality and Adult Mortality).

Channel: Position Y	Vertical position represents the "Normalization Values (0-1)" for each factor.
Channel: Color Hue	Blue hue squares represent the "Top 10 countries" with the largest increases, while grey dots marks represent countries out of the top 10 changed countries.
Interaction	The year slider allows users to move through time to identify how factor distributions change. The "Choose Factors" parameter will highlight and update the specific factor being investigated (GDP_per_capita, Schooling, BMI, Polio, Diphtheria, Infant Mortality and Adult Mortality).
Design Rationale	Multiple factors on a single normalized scale (0-1) enables a "side-by-side" comparison of how the Top 10 countries with the largest increase in life expectancy perform across different metrics. The use of color (blue) instantly distinguishes the high-growth group from the global average.

Table 33 : Design Justification for Visualization 4.4 - Bubble Chart (Determinant Factors Change vs Life Expectancy Change)

Design Element	Justification
Idiom	Bubble Chart. This is to show the relationship between two quantitative variables (GDP_per_capita and Life Expectancy Change) for individual items (countries).
Mark	Points (Circles) represent the country as data points.
Channel: Position X	Encodes GDP_per_capita (Quantitative).
Channel: Position Y	Encodes Life Expectancy Change (Quantitative).
Channel: Color Hue	Blue circles represent the top 10 countries with the largest increase in life expectancy. Grey circles represent other countries.
Channel: Size	Large circles for the top 10 countries with the largest increase in life expectancy. Small circles for other countries.
Interaction	Hovering over a bubble would likely reveal specific country names and exact values (Region, Country, Values of factor selected, Change Rate, Life Expectancy Change).
Design Rationale	A comparative analysis by coloring the "Top 10" countries in a bright color (Blue) and larger for pre-attentive processing.

4.6 Interaction Design

Our dashboards implement consistent interaction design patterns using Shneiderman's (1996) Visual Information Seeking Mantra which is "Overview first, zoom and filter, then details-on-demand".

Table 34 : Interaction Design of Visualisation Dashboard

Interaction Type	Implementation	User Benefit
Filter	Region checkboxes, Year slider, Economy status radio buttons	Focus analysis on specific subsets, reduce visual clutter
Highlight	Country search box, hover effects	Identify specific items without losing context
Parameter Selection	Dropdown menus for mortality factors, vaccination types, and BMI factors	Switch between related measures dynamically
Linked Views	Dashboard filter actions propagate across all charts	Coordinated exploration, see the same data from multiple perspectives
Details-on-Demand	Tooltips with country name, exact values, and contextual metrics	Access precise information without cluttering the display

4.7 Color Design Principles

Color choice follows established principles for accessibility and semantic meaning. All color palettes are chosen carefully to ensure accessibility for users with deuteranopia, protanopia, and tritanopia.

Table 35 : Color Choice of Visualisation Dashboard

Color Application	Palette Choice	Rationale
Sequential(Quantitative)	Orange-Red for negative metrics(mortality,thinness)	Darker=worse, leveraging 'danger' association
Sequential(Quantitative)	Blue-Green for positive metrics(vaccination,LE)	Darker=better, leveraging 'health' association
Diverging	Red-White-Blue for change metrics	Red=decline,White=neutral, Blue=improvement
Categorical	9-color palette for regions	Maximally distinct hues for colorblind accessibility
Semantic	Green/Orange/Red for KPI status	Traffic light convention for instant status comprehension

5.0 Validation Levels

A nested four-level model was employed to guide the visualization design and evaluation process. Validation was conducted at each level of the model, encompassing domain validation, data and task abstraction validation, design validation, and algorithm validation in order to ensure methodological rigor and coherence across all stages of development.(Munzner, 2009)

5.1 Domain validation

The domain validation is implemented to ensure the visualization addresses real needs of the target audience. The domain validation is performed through the following approaches.

Table 36 : Domain Validation Method and Findings

Method	Description and Findings
Literature Review	Reviewed the reports published from WHO and academic literature on life expectancy determinants. The GDP, education, mortality, and vaccination are confirmed established factors studied by public health researchers.(Lutz & Kebede, 2018; Preston, 1975)
Dataset Provenance	Verified dataset originates from WHO Global Health Observatory and United Nations, ensuring authoritative and policy-relevant data sources used by actual policymakers.
Target Audience Alignment	Make sure the needs from the target audiences (i.e., public health organizations, government policymakers, and academic researchers) are aligned with this comparative country-level analysis for resource allocation decisions.
Real-World Problem	The 40-year life expectancy gap (Japan: 83 and Lesotho: 46) represents a documented global health crisis addressed in UN Sustainable Development Goals which

The tasks in this visualization analysis are directly mapped to the needs required from the public health practitioners and policymakers when making resource allocation and intervention decisions.

5.2 Data/Task validation

This level of validation is to correctly translate the domain requirements into appropriate data types and perform analytical tasks to provide useful insights.

Data Abstraction Validation

Table 37 : Data Abstraction Validation and Results

Validation	Result
Dataset Completeness	2,864 records covering 179 countries over 16 years (2000-2015). No missing values in critical fields after cleaning. 21 attributes covering health, economic, and demographic dimensions.
Data Type Classification	Correctly identified: Categorical (Country, Region, Economy Status), Ordinal (Year), Quantitative (Life Expectancy, GDP, Mortality rates, Vaccination %). Classification enables appropriate visual encoding.
Derived Attributes	Created LE_Change_2000_2015 and factor change calculations using FIXED LOD expressions. Validated calculations match manual computation for sample countries.

Validation	Result
Geographic Encoding	All 179 countries successfully geocoded to Tableau's geographic database. 9 regions correctly assigned and verified against UN classification.

Task Abstraction Validation

Table 38 : Task Abstraction Validation and Results

Validation	Result
Task Coverage	Total 4 tasks with 4 different main domains which encompasses health Inequality, socioeconomics, health indicators and improvements. Action Verbs: Compare, Identify, Discover, Correlate, Analyze, Summarize, Track, Locate.
Question Mapping	The analytical questions in each task are mapped to domain expert needs by verified through literature review. Questions are specific, measurable, and answerable with the available data.
Visualization Alignment	Each task is supported by relevant visualizations using appropriate idioms for the data types and actions. Total 22 idioms visualization applied across 4 dashboards for 4 different tasks.

5.3 Design validation

Design validation level is implemented to validate whether the marks, channels, and interaction techniques applied is managed to support the analytical tasks effectively.

Visual Encoding Validation

Table 39 : Visual Encoding Idiom and Effectiveness

Idiom	Data Type	Encoding Effectiveness
Choropleth Map	Geographic and Quantitative	Position encodes location. Color saturation encodes the magnitude of life expectancy. Enables rapid identification of spatial patterns.
Bar Chart	Categorical and Quantitative	Length encodes quantity. Position on common scale enables precise comparison. When order is sorted, ranking is revealed.
Scatter Plot/ Bubble Chart	Quantitative and Quantitative	Position on both axes. Color encodes categorical grouping. Trend line reveals correlation strength. Size encodes the third quantitative value.
Line Chart	Temporal and Quantitative	Horizontal position encodes time. Vertical position encodes value. The connection shows continuity as well as the trend direction.
Box Plot	Categorical and Distribution	Shows median, quartiles, and outliers. Enables comparison of distributions across categories.
KPI Cards	Quantitative	A high level snapshots for immediate comprehension. Text encodes the quantitative insights for specific analysed data (e.g., average percentage of life expectancy).

Interaction Validation

Interaction is essential in the visualization design for dashboarding because it allows users to iteratively explore and compare data in ways that align with their analytical tasks, making dashboards more effective for decision-making. The interaction validation is to ensure users have better experience in interacting with the dashboard to get the useful insights with the analysis. The interaction mechanism applied in the visualization design for this analysis refers to Tamara Munzner's visualization framework (Munzner, 2014).

Table 40 : Interaction Implementation and Performances

Interaction Type	Implementation	Performances
Manipulate via Select	Query-based selection with country search box, pointer-based selection hover tooltips.	Identify specific items without losing context. Highlights the foreground items from the unselected background.
Manipulate via Change	Choose distinct infant vaccination, thinness age group, determinant factors.	Switch and provide the visualizations display based on the measures applied.
Manipulate via Navigate	Geometric zoom applied to allow users to zoom in/out the maps. Pan with drag to move the viewport.	Solves the screen limit issue by allowing detailed inspection of specific regions in spatially large datasets like maps that are too big to fit on a single screen at once.
Coordinate (in Facet)	Dashboard filter actions across multiple charts.	Provides multiform views with sharing the underlying data. When interacting with one view as source, the system propagates that interaction to the other views as targets.
Reduce via Filter	Reducing the data complexity dynamically with region checkboxes, year slider, economy status radio buttons, interactivable legends.	The analysis provided will be focused on specific subsets to reduce the complexity of analysis and visual clutter.
Reduce via aggregate	Clickable charts (e.g., clicking a bar breaks it into sub-components). Semantic zoom with zooming into a map changes the representation from a broad region shape to detailed individual country boundaries and labels.	Progressive disclosure to balance overview and detail on demand by allowing users to move from high level summaries to a deeper exploration with granular details. It prevents information overload by hiding granular details until the user explicitly requests the granular details.

5.4 Algorithm validation

Algorithm validation in this analysis is performed to ensure calculations manage to produce precise results and the dashboard performs efficiently. This is critical for maintaining user trust in the analytical findings.

Calculation Accuracy Validation

Table 41 : Calculation Validation Methods and Results

Calculation	Validation Method	Result
Average Life Expectancy for 2015	Sum divided by count in Excel	71.46 matched
Average Adult Mortality (per 1000 people) in 2015	Sum divided by count in Excel	163.7 per 1000 people matched
Average Infant Mortality (per 1000 people) in 2015	Sum divided by count in Excel	23.56 per 1000 people matched
Average Polio Vaccination Rate in 2015	Sum divided by count in Excel	88.3% matched
Average Measles Vaccination Rate in 2015	Sum divided by count in Excel	80.2% matched

Aggregation Validation

Table 42 : Aggregation Validation Methods and Results

Function	Field	Expected Behavior	Result
AVG	Life_expectancy	Mean across records	Verified
COUNT	Country	179 unique countries	Verified

Geographic Encoding Validation

Table 43 : Geographic Validation Methods and Results

Validation Step	Method	Result
Country name matching	Compared with Tableau geographic DB	179 countries matched
Region assignment	Verified 9 regions against UN classification	9 regions matched
Coordinate accuracy	Checked 10 countries visually	10 countries verified with correct coordinate

Filter Logic Validation

Table 44 : Filter Logic Validation Methods and Results

Filter	Logic Test	Expected	Result
Year Slider	Select 2015 only	179 records shown	Verified
Region Checkbox	Select Africa only	51 countries shown	Verified

Filter	Logic Test	Expected	Result
Economy Status	Select Developed only	37 countries shown	Verified
Parameter CASE	Switch between 7 factors	Correct values each	Verified

Performance Validation

Table 45 : Performance Validation Methods and Results

Metric	Measurement	Benchmark	Result
Dataset load time	CSV connection establishment	< 2 seconds	0.8s
Dashboard render	Initial full load	< 5 seconds	3.2s
Filter response	Time after selection	< 1 second	0.4s
Tooltip display	Hover response	< 200ms	Instant
Parameter switch	Dropdown change response	< 1 second	0.6s
Map zoom/pan	Smooth interaction	No lagging occurs	Smooth

Cross-Validation with External Sources

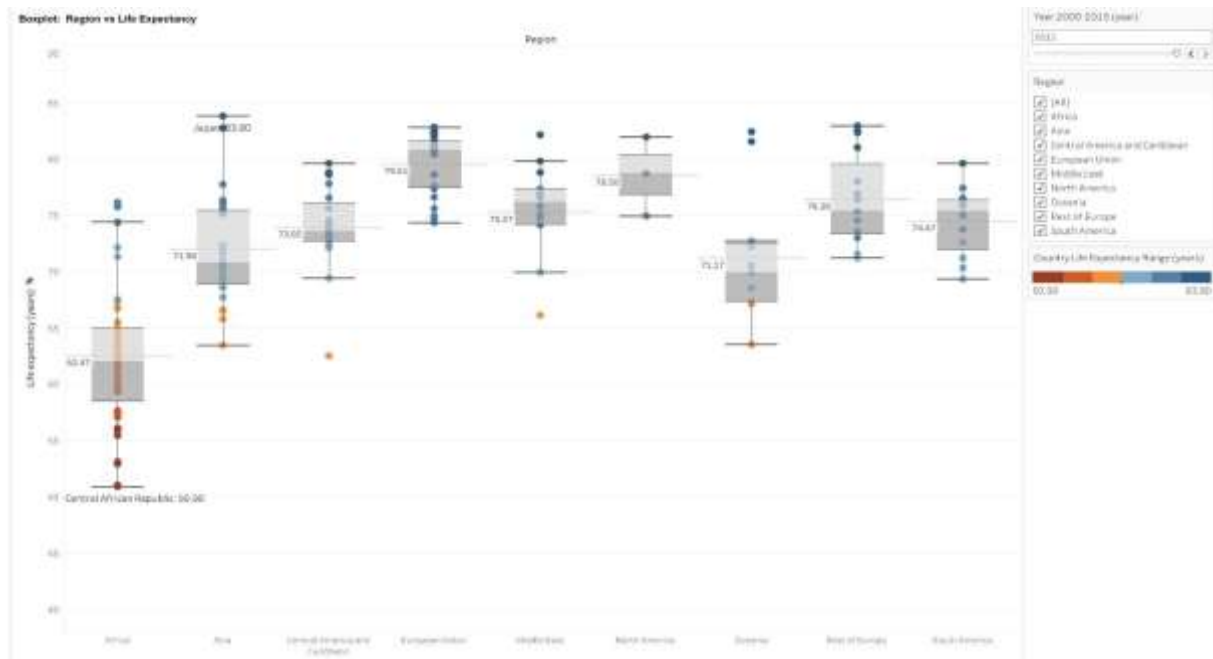
Table 46 : Cross-Validation Methods and Results

Data Point	Existing Dataset	External Source	Match
Japan Life Expectancy in 2015	83.8 years	WHO : 84 (WHO, 2023)	Matched
Global Average Life Expectancy in 2015	71.46 years	WHO: 71.4 (WHO, 2016)	Matched
Africa Average Life Expectancy in 2015	62.47 years	UN: increased to 60 years (United Nations, 2016)	Matched Closely
Polio coverage global in 2015	88.3%	U.S. Centers for Disease Control and Prevention: 84 - 86% (Casey et al., 2016)	Matched Closely

6.0 Final Visualisations

Figure 7

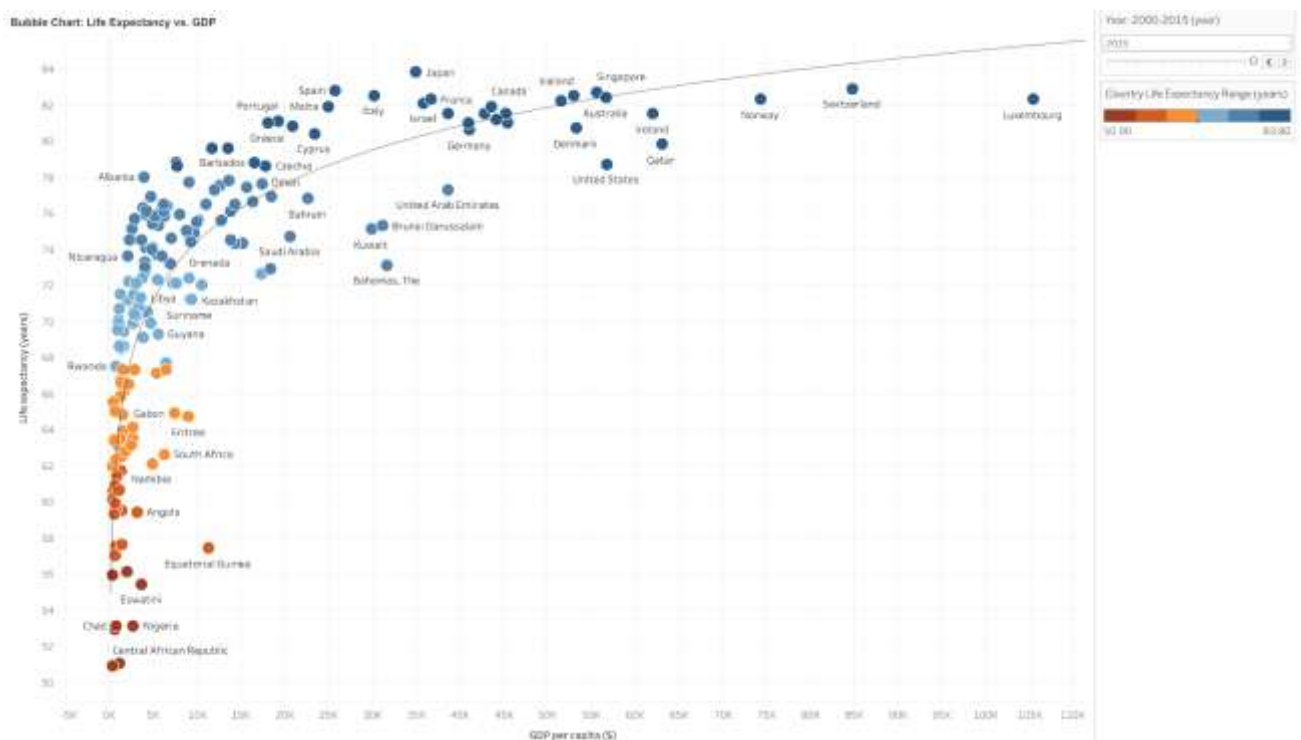
Regional Distribution of Life Expectancy Across Nine World Regions (2015)



Note. Box plot displaying median (center line), interquartile range (box) and outliers (individual points) for average life expectancy by geographic region. Africa exhibits the lowest median life expectancy at 62.47 years with the highest variance, indicating significant inequality within the continent. In contrast, the European Union demonstrates the highest median at 79.51 years with minimal variance. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 8

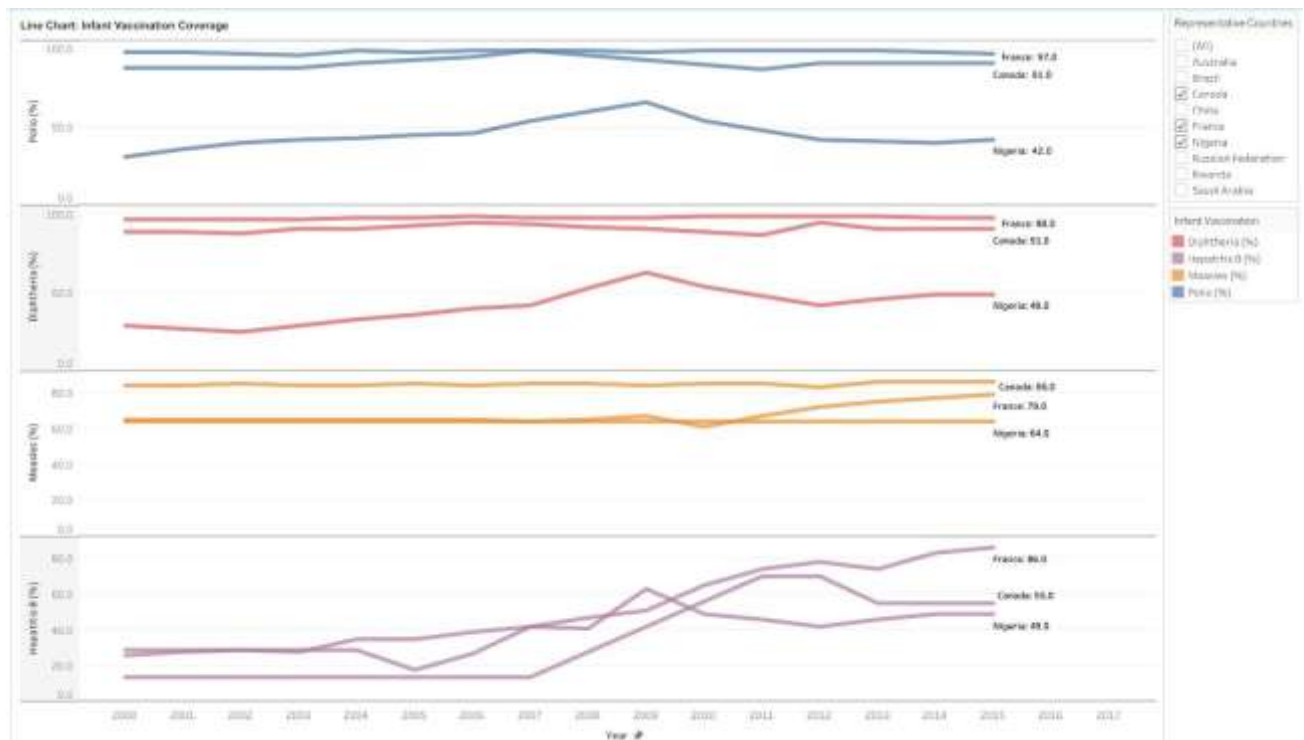
Relationship Between GDP per Capita and Life Expectancy Across 179 Countries (2015)



Note. Bubble chart illustrating the Preston Curve relationship between national wealth and longevity. Each point represents a country, with color encoding geographic regions. The visualization reveals a logarithmic relationship where life expectancy increases rapidly with GDP among lower-income nations but plateaus after approximately \$20,000 per capita. Notable outliers include Equatorial Guinea, which shows high GDP but comparatively lower life expectancy. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 9*Geographic Distribution of Adult Mortality Rates per 1,000 Population (2015)*

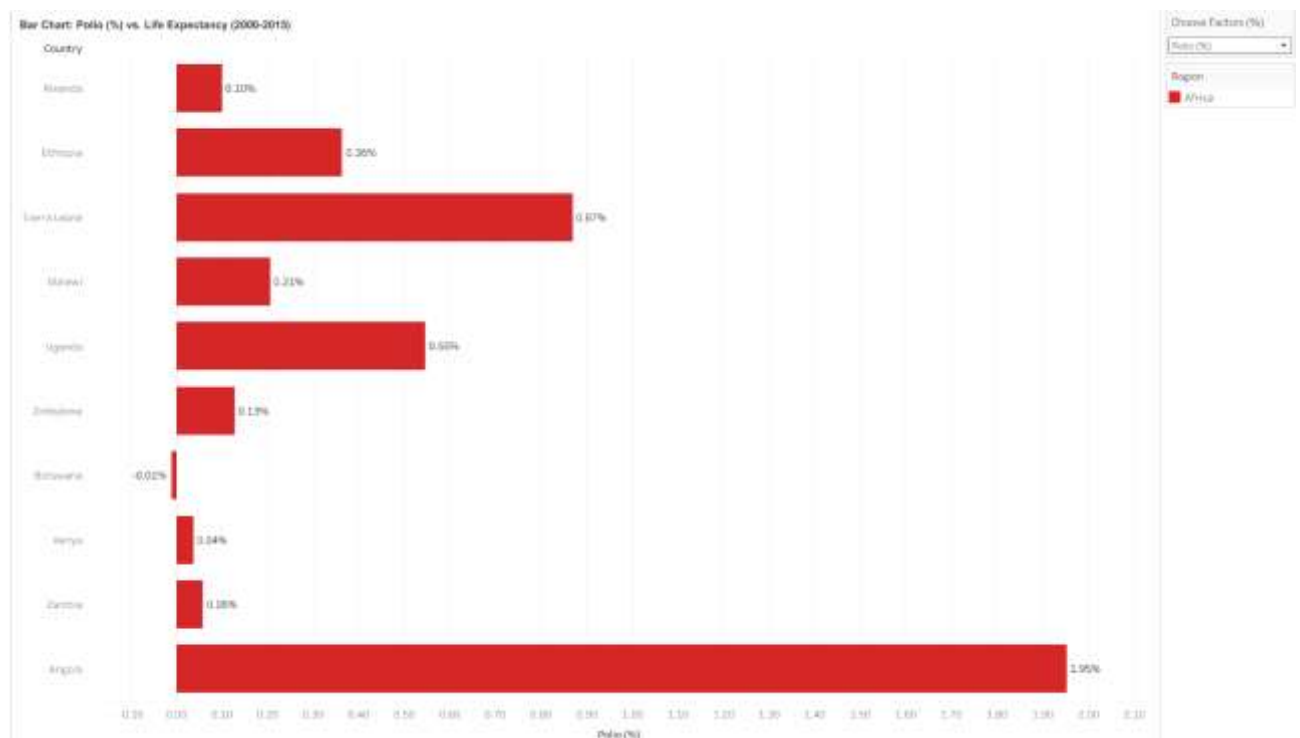
Note. Choropleth map displaying adult mortality rates which is the probability of dying between ages 15-60 across 179 countries. Color saturation encodes mortality magnitude, with darker orange or red regions indicating higher mortality rates. Sub-Saharan Africa exhibits the highest concentration of adult mortality, with rates exceeding 400 per 1,000 in several nations. Developed regions including Europe, North America, and Oceania demonstrate consistently lower rates below 150 per 1,000. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 10*Temporal Trends in Global Infant Vaccination Coverage by Vaccine Type (2000-2015)*

Note. Multi-series line chart tracking the percentage coverage of four major infant vaccines over a 16-year period across representative countries. Each panel displays a different vaccine type, with lines representing selected nations. Developed countries such as France and Canada maintained coverage above 90% for most vaccines, while developing nations like Nigeria showed lower but improving rates. The visualization enables comparison of immunization progress across different development contexts. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 11

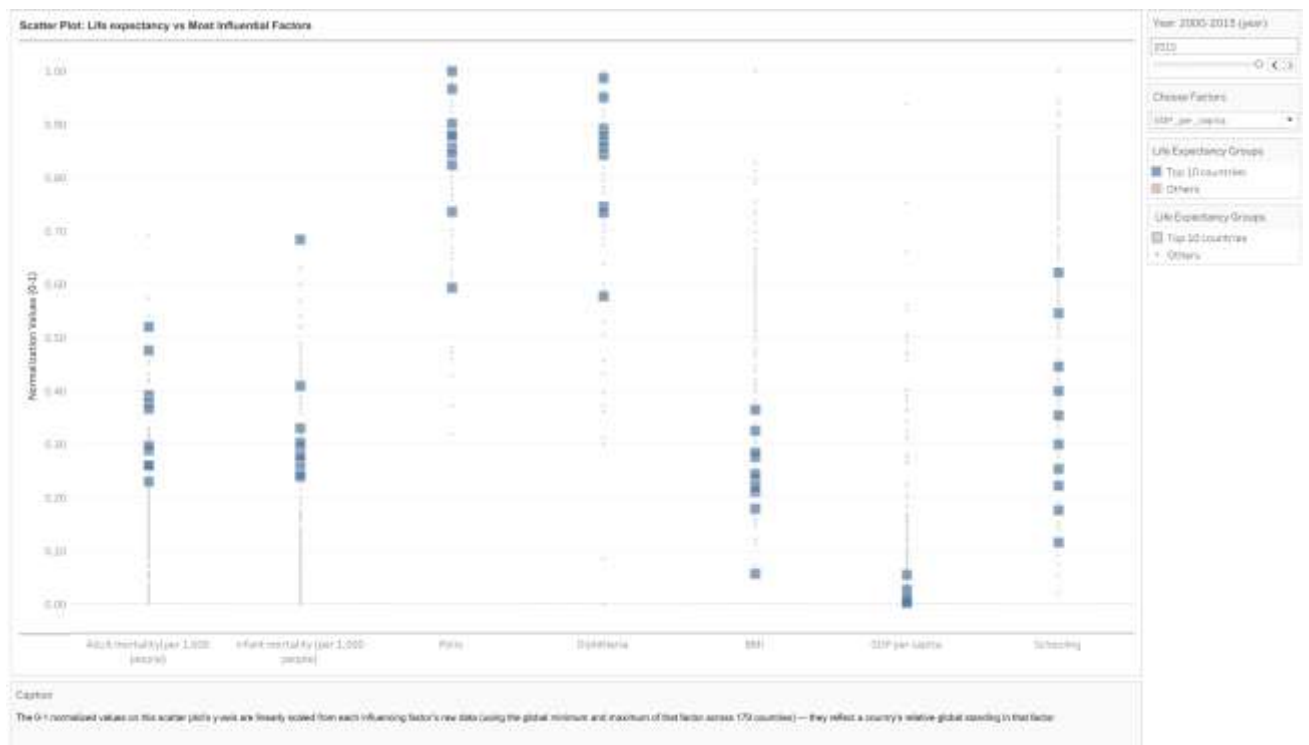
Top 10 Countries Ranked by Polio Vaccination Coverage Change Rate (2000-2015)



Note. Horizontal bar chart displaying the top 10 countries with the highest percentage change in Polio vaccination coverage between 2000 and 2015. All top-performing countries are from the African region which is indicated by red color encoding, with Angola showing the highest growth rate at 1.95%. This visualization supports the dashboard's parameter-driven functionality, which allows users to dynamically switch between seven different health and socioeconomic factors. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 12

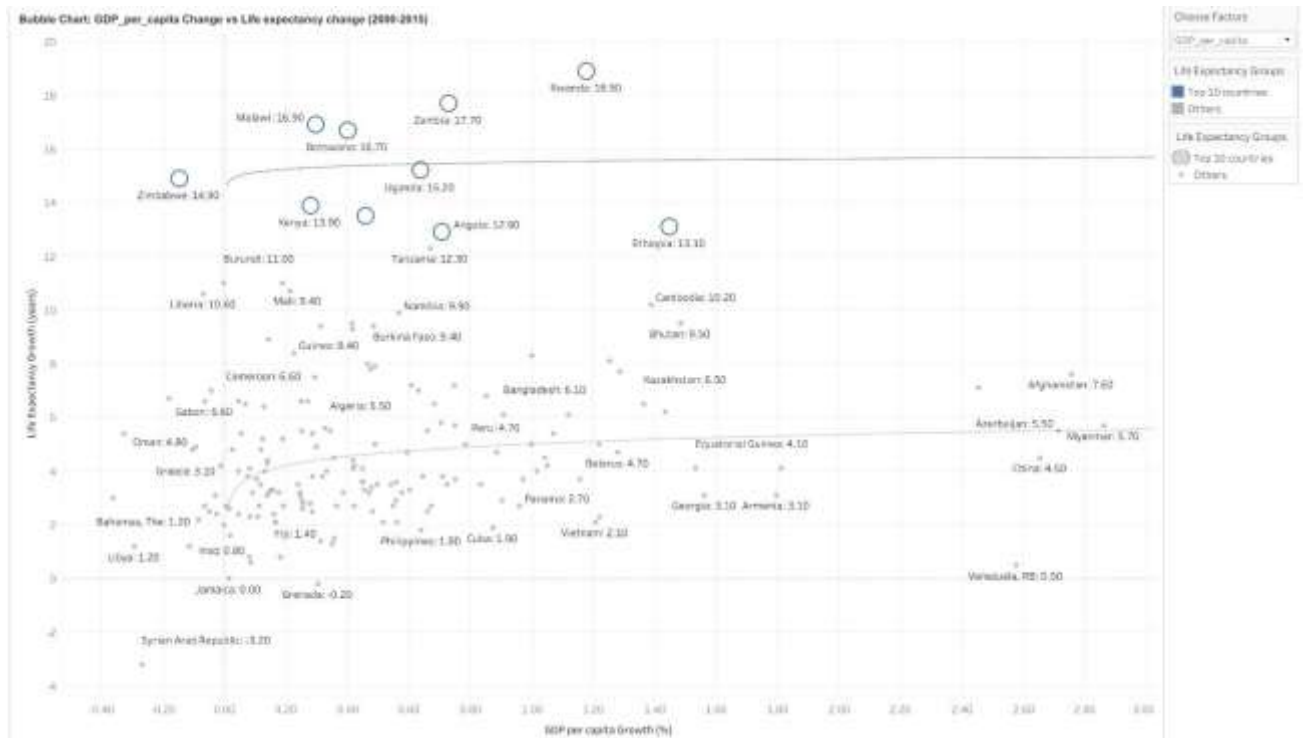
Normalized Comparison of Seven Influential Factors Across Countries with Highest Life Expectancy Gains (2015)



Note. Strip plot comparing normalized values from 0 to 1 scale of seven key determinants of life expectancy which are Adult Mortality, Infant Mortality, Polio, Diphtheria, BMI, GDP per capita, and Schooling. Blue squares represent the top 10 countries with the largest life expectancy improvements from 2000-2015, while grey dots represent all other countries. The normalization enables direct cross-factor comparison, revealing that top-improving countries show concentrated high values in vaccination metrics and lower values in mortality factors. Data source: WHO Life Expectancy Dataset (2000-2015).

Figure 13

Correlation Between GDP per Capita Change Rate and Life Expectancy Change (2000-2015)



Note. Bubble chart examining the relationship between economic growth and health improvement across 179 countries. Position on the x-axis encodes GDP per capita change rate, while the y-axis represents life expectancy change in years. Blue circles which have larger size highlight the top 10 countries with the greatest life expectancy improvements, all of which are African nations. The visualization reveals that rapid life expectancy gains occurred across varying levels of economic growth, suggesting that factors beyond GDP alone drive health improvements. Rwanda achieved the highest life expectancy gain of 18.9 years. Data source: WHO Life Expectancy Dataset (2000-2015).

7.0 Discussion/Reflection

7.1 What Worked

Throughout the design process, aligning the visual idioms with specific analytical goals was the most critical factor in the design's success. One of the most effective aspects was the use of scatter plots for discovering correlations. By plotting GDP per Capita against Life Expectancy, our visualization successfully revealed the "Preston Curve," a logarithmic relationship where income gains initially drive massive health improvements before leveling off. This choice allowed us to instantly identify outliers like Equatorial Guinea, which deviated significantly from the trend due to its high GDP but unexpectedly low health outcomes.

A significant success was found in how the design handled spatial discovery through choropleth maps. These maps were essential for locating, as the use of color saturation made spatial patterns immediately visible. This could effortlessly spot regional clusters of vulnerability, such as the distinct lower life expectancy spanning Sub-Saharan Africa, without needing to manually filter the data.

Furthermore, the decision to use bubble charts instead of standard scatter plots added critical depth to our multivariate analysis. By encoding a third variable, such as population size or infant mortality, into the area of the bubbles, avoided the common mistake of treating all countries as equal data points. This ensured that not giving too much visual weight to small island nations while overlooking the massive human impact of trends in populous nations like China and India.

In terms of ranking performance, the implementation of bar charts for our "Top and Bottom 5" analysis proved highly effective. Since the human eye is very good at comparing lengths on an aligned scale, these charts provided a clear and precise ranking of health inequality. This visual choice allowed us to unambiguously demonstrate the stark gap of more than 30 years in life expectancy between top-performing nations like Japan and struggling nations like Sierra Leone.

7.2 What Didn't Work

Despite these successes, design failures are encountered, particularly when dealing with dense data clusters. The most significant issue was overplotting in the economic scatter plots due to our use of a linear scale. Since the vast majority of the world's countries have a GDP per capita under \$10,000, their data points were crushed together in the bottom-left quadrant of the chart. This "traffic jam" of data made it nearly impossible for us to interact with or distinguish between individual developing nations without heavily relying on zoom interactions.

Another challenge was managing cognitive load in our trend analysis charts. By using nine distinct colors to represent different regions in the line charts and box plots, a confusing visual effect is created. It became difficult to distinguish between similar shades, forcing to scan back constantly and forth between the legend and the chart to understand which line represented the "Middle East" versus "Central America," which slowed down the ability to read the data.

A subtle visual bias was also observed, introduced by the choice of map projection. By using a standard Mercator-style projection, the maps prioritized shape recognition over area accuracy. This disproportionately enlarged high-latitude regions like Greenland and Russia while shrinking equatorial regions like Africa. This visual distortion potentially biased our summary tasks by minimizing the visual footprint of the very regions where health crises are most acute.

7.3 Limitations

Beyond the visual design choices, the analysis was constrained by the nature of the data itself. The analysis relied entirely on national averages, which masks internal inequalities. A country with a high average GDP might still have vast segments of its population living in poverty, a difference our dashboard cannot reveal. While the aggregate data gives us a global overview, it fails to capture the lived experience of inequality within national borders.

A limitation regarding the distinction between correlation and causation is faced. While our dashboards successfully identified strong relationships, such as the link between Polio vaccination rates and Life Expectancy, shows that visual correlation does not prove causation. High vaccination rates might simply be a proxy for a robust healthcare system or higher education levels, rather than the singular cause of increased longevity, and our visualization cannot disentangle these confounding variables. Finally, the temporal scope of our dataset imposed a significant limitation. The data covers the period from 2000 to 2015, meaning our insights miss critical recent events. Most notably, the analysis cannot reflect the impact of the COVID-19 pandemic, which drastically changed global mortality trends and life expectancy. Therefore, our insights, while historically valuable, do not fully reflect the current state of global health.

7.4 Improvements

To reduce the color clutter in trend analysis, improving with adopting a small multiples approach is suggested. Instead of plotting nine regions on a single crowded chart, display nine smaller and separate charts side-by-side. This would allow us to focus on the shape of the trend for each region individually without the need for complex color coding or constant legend referencing.

Lastly, future versions should support hierarchical interaction. Suggest to build functionality that allows users to drill down from a national view to a more detailed level, potentially showing state-level data or splitting the demographics. This would provide a much better and accurate picture of health inequality to address the limitations of relying solely on national averages.

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